

Mahmoud Elboghday . Jarvis Consulting

I'm a Business Systems Analyst moving into machine learning engineering through the Jarvis ML stream, with 3+ years of experience turning operational complexity into measurable business outcomes through requirements elicitation, BRD and user-story authoring, stakeholder management, and Agile delivery. Grounded in business fundamentals from a Hospitality Management degree and sharpened through an Information Systems Engineering diploma alongside hands-on delivery, from authoring a 13-section Business Requirements Document for an RBC Global Asset Management reporting dashboard and building an explainable credit scoring model, to serving as the primary data analyst at Desert Suspension (an authorized 4x4 vehicle equipment dealer operating across Saudi Arabia) across Sales, Marketing, and executive leadership, delivering end-to-end data solutions that drove significant improvements in reporting efficiency, inventory accuracy, and profitability.

Skills

Proficient: SQL/RDBMS (PostgreSQL, SQL Server, Oracle), Requirements Elicitation & BRD/User Stories, Stakeholder Management, Agile/Scrum (planning, standups, retros), Python, Data Analysis (Pandas, Numpy), Power BI/DAX, Machine Learning (scikit-learn, tensorflow, PyTorch), GCP/AWS, Git/GitHub

Competent: UML/System Design, Java/Spring Boot, SDLC, Docker, ETL Pipeline Development, Model Explainability & Monitoring (SHAP, PSI), Linux/Bash, Spark/Databricks

Familiar: Azure, Tableau, Kubernetes, React, UI/UX

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_MahmoudElboghday

Home Credit Default Risk - Credit Scoring Model. [GitHub]: Built an end-to-end credit scoring model on the Kaggle Home Credit dataset (307,511 loan applications, 122 columns) joined with credit bureau history, to predict which applicants will have payment difficulties. Fixed placeholder codes and data errors, kept missing-value indicators as features, and engineered affordability, employment and bureau features from one shared module used by both the research notebooks and the production code. Compared Logistic Regression, Random Forest and Gradient Boosting with 5-fold cross-validation and randomized search tuning; the tuned Gradient Boosting model reached 0.771 AUROC (Gini 0.54, KS 0.41) on a 61,503-application holdout, above the 0.72 target. Explained predictions with SHAP and adverse action reasons, set up PSI drift monitoring, kept sex out of the model inputs, and shipped a scikit-learn pipeline that scores raw application rows (0.768 AUROC) with SR 11-7 style model documentation.

RBC GAM Monthly Reporting Dashboard - Business Requirements Document. [GitHub]: Led requirements elicitation for an automated Monthly Performance Reporting dashboard for RBC Global Asset Management, replacing a manual ~25-hour, five-system process that had produced an executive-facing data error. Ran a structured interview with the sponsoring Managing Director and routed current-state and technical questions to the data analyst and development lead, then authored a complete 13-section Business Requirements Document with testable functional and non-functional requirements (Given/When/Then), MoSCoW prioritisation, a RACI matrix, dependency and risk registers, and a burn-down tracker. Managed scope through formal change control (deferring a CFO real-time-feed request to a later phase), logged unresolved dependencies with owners and recommended options (including a broken vendor API and a direct-database compliance review), and delivered a 10-minute executive briefing recommending sign-off within the governance cycle.

Linux Cluster Resource Monitoring System. [GitHub]: Designed and deployed a Linux cluster monitoring agent on Rocky Linux 9 (GCP) to give administrators real-time visibility into CPU, memory, and disk usage across distributed nodes, supporting informed decisions around resource allocation and capacity planning. Built two Bash scripts to collect hardware specs and live usage metrics, persisting data into a centralized PostgreSQL instance containerized via Docker. Automated metric collection with crontab and validated end-to-end pipeline accuracy through SQL-based analysis.

Retail Stock Trading Platform Software Requirements Specification. [GitHub]: Conducted stakeholder elicitation to author a comprehensive Software Requirements Document (SRD) for a full-stack web and mobile retail self-directed trading application commissioned by a Canadian bank. Structured, clear functional boundaries (FRs) managing complex, atomic multi-account portfolios (TFSA, RRSP, Cash Accounts), strict buy/sell trade constraints, and 5-minute interval third-party Market Data API integration flows. Engineered robust non-functional requirements targeting high security, system availability, and data integrity parameters for an expected launch volume of 150,000 to 200,000 active clients.

Retail Data Analytics & Customer Segmentation Pipeline. [GitHub]: Developed a Python-based Proof of Concept analytics pipeline inside Docker to convert 1.07 million rows of raw historical transaction data from a PostgreSQL database

into actionable marketing intelligence. Built clean, top-down Jupyter Notebook pipelines executing data wrangling, string normalization, snake_case table mappings, and strict split tracking for placed versus cancelled orders to insulate revenue reporting. Engineered custom Recency, Frequency, and Monetary (RFM) features to categorize the customer base into distinct, operational clusters (e.g., Champions, At Risk, Hibernating). Delivered a high-level executive presentation featuring professional Seaborn and Matplotlib visualizations mapping month-over-month growth patterns and demand peaks to guide seasonal stock optimization.

Databricks Lakehouse ETL & Streaming Analytics Pipelines. [GitHub]: Built two independent PySpark pipelines on Databricks to contrast batch and streaming lakehouse design patterns. Engineered a bronze/silver/gold medallion ETL pipeline over a 13.3-million-row credit card transaction dataset sourced from Azure Data Lake Storage Gen2, cleaning and typing raw records in Silver and rolling them into Gold fraud-analytics tables (fraud rate by day, time of day, merchant category, and user) queryable through Unity Catalog. Built a companion Delta Live Tables pipeline that ingests live market data from the Alpha Vantage REST API into a Databricks Volumes landing zone, then declares an incremental bronze/silver/gold table chain with Auto Loader (cloudFiles) for streaming, checkpointed processing. Documented both pipelines with an architecture diagram and dataset/table-level breakdowns for handoff.

Highlighted Projects

Law Enforcement BI and Resource Optimization Pipeline. [GitHub]: Built an end-to-end crime analytics platform on 1.23M+ Chicago Police Department incidents (2021-2025) using a SQL Server star schema (4 dimension tables, 1 fact table) and a Python ETL pipeline maintaining 96% data completeness across critical fields. Delivered descriptive, diagnostic, predictive, and prescriptive analytics across the full stack. Trained three ML models: a Random Forest crime-type classifier (88% accuracy, top features: hour of day and severity level), a Gradient Boosting volume forecaster (R-squared 0.62, RMSE 50.66 crimes/day, MAPE 21.23%), and a DBSCAN geospatial hotspot detector (epsilon 0.01 approx. 1km radius, min samples 20) that identified 15 high-risk clusters accounting for 35% of all incidents.

Professional Experiences

Machine Learning / Generative AI Developer (Co-op), Sheridan College - Centre for Applied AI (CAAI) (05/2026-present): Translate high-level academic and enterprise research objectives into detailed software specifications, feasibility assessments, and software architecture designs. Design and prototype deep machine learning approaches utilizing encoder-decoder neural network architectures and context-aware patterns to resolve indoor localization resolution and signal reliability barriers. Evaluate third-party models and open-source frameworks against strict performance requirements to champion data-driven build-versus-buy infrastructure decisions.

Business Systems Analyst, Jarvis (04/2026-present): Delivering BSA and data projects in an agile, sprint-based environment with structured code reviews and daily standups. Authored a 13-section Business Requirements Document for an RBC Global Asset Management reporting dashboard, leading stakeholder elicitation, change control, and risk and dependency management. Built a Linux cluster monitoring agent using Bash, PostgreSQL, and Docker to automate resource metric collection across a distributed environment. Now in the Jarvis Machine Learning stream, where I built an explainable credit scoring model (tuned Gradient Boosting, 0.771 AUROC) with SHAP explanations, PSI monitoring and a production scikit-learn pipeline.

Business Intelligence Analyst, Desert Suspension (08/2024-03/2026): Owned end-to-end portfolio analytics and operational reporting across three international facilities, serving as the primary liaison between Sales, Marketing, and executive leadership. Automated ETL workflows using Python and SQL, reducing manual reporting cycles by 75% and eliminating cross-departmental data silos. Applied statistical trend analysis to inventory data, achieving 85% forecasting accuracy and contributing to a 17% increase in net profit. Governed a centralized data warehouse maintaining integrity and usability standards across all pipelines. Produced audit-ready technical and business documentation ensuring reproducibility and knowledge continuity across reporting deliverables. Operated in an agile workflow, partnering with cross-functional stakeholders to align data outputs with strategic business priorities.

Data Analyst, Upwork (02/2023-01/2024): Delivered freelance data analytics engagements across multiple clients, producing over 10 Power BI dashboards with real-time KPI tracking that accelerated decision-making speed by 40% for 15+ stakeholders. Built automated SQL ETL pipelines consolidating multiple data sources at 98% accuracy, and designed A/B tests using Python statistical methods that drove a 15% improvement in customer engagement metrics.

Education

Sheridan College (2025-present), Advanced Diploma, Information Systems Engineering

University of Alexandria (2015-2019), Bachelor of Science, Hotel Management and Hospitality

Miscellaneous

- IBM Data Science Professional Certificate
- DataCamp Power BI Data Analyst Certificate
- PwC Data Analytics Job Simulation
- Accenture Data Analytics Job Simulation
- Competitive athlete across multiple disciplines: started as a competitive swimmer, progressed to men's physique bodybuilding, and competed in CrossFit team events across the Middle East (Saudi Arabia and Bahrain).
- Currently focused on staying active and healthy as a new father, training to keep up with my daughter as she grows.
- Content creator with over 300k followers covering Canadian immigration topics and the licensing process for foreign-trained professionals, interviewing licensed professionals who completed the journey and now give back to the community.